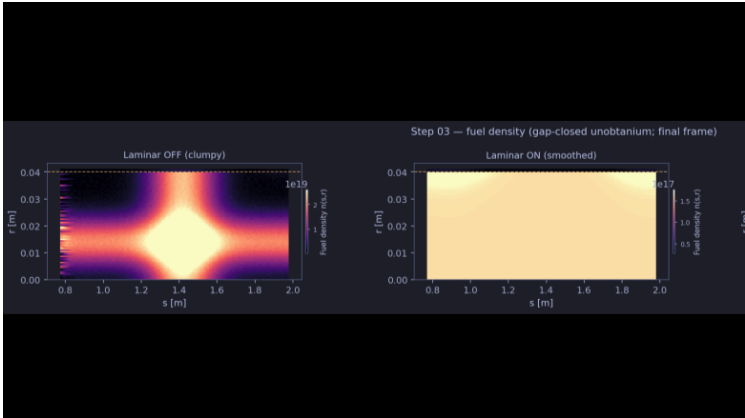


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# Orbitron Direct Cycle p-11B 3.5MW Turbojet In Silico Benchmark (REVISED 5/27)

Lars Warren Ericson

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In-silico benchmark — 2026-05-28 00:26

## Introduction

This report is inspired by a fusion device called the Orbitron developed by [Avalanche Energy](#) . It explores the feasibility of an air-breathing jet propulsion system powered directly by a proton-boron (p-<sup>11</sup>B) Orbitron-style fusion reactor designed to produce approximately 3.5 megawatts (MW) of total raw power.

Here  $\langle\sigma v\rangle$  denotes the **fusion reactivity**: the fusion cross section  $\sigma(E)$  multiplied by ion relative speed  $v$ , then averaged over the ion velocity distribution at the operating ion temperature. In this notation,  $E$  is center-of-mass collision energy,  $n_1$  and  $n_2$  are the reacting-ion number densities (for p-<sup>11</sup>B, proton and boron-ion density), and  $R$  is volumetric fusion reaction rate (reactions per m<sup>3</sup> per s). The model scaling is  $R \propto n_1 n_2 \langle\sigma v\rangle$ , so  $\langle\sigma v\rangle$  is the key bridge from plasma conditions to predicted fusion power. In this benchmark, changing from the design-calibrated branch to the literature branch changes  $\langle\sigma v\rangle$  by about three orders of magnitude near the operating point, which is why the same geometry and fueling can move from MW-class closure to a strong shortfall.

HTS means **high-temperature superconductor** — here, a **liquid-methane-cooled solenoid outside a vacuum cryostat** that projects 2 T into the bore (U3). It is **not** the surface that compressed air washes; air flows in a **hot annulus inside the magnet radius** over the **first wall** (see **Benchmark Methodology — Radial thermal zoning**). p-<sup>11</sup>B does **not** heat the engine via neutrons; the wall catches **alphas and X-rays**.

We report **three scenarios** (definitions and rules in **Benchmark Methodology** below):

1. **(a) Pretend — design target:** design-calibrated  $\langle\sigma v\rangle$ , 600 kV class, unity unobtainium. The proof chain (steps 0–8) runs on this path. Level-1 plant closure is **not** measured fusion yield.
2. **(b) Today — COTS (Commercial Off The Shelf) + experiment:** literature  $\langle\sigma v\rangle$ , Avalanche-class **300 kV**, same pad fueling as (a), wall and HTS limits at published values. No tuning to recover MW.
3. **(c) Minimum — stress inverse:** literature  $\langle\sigma v\rangle$  with optimizer free to raise fusion\_reactivity\_scale ( $\sim 10^3\times$ ) to approach target; margin inverse checks back-solve  $\approx$  (a).

The physical geometry was modeled using CadQuery and Blender. **WarpX PIC (step 01)** validates electron loading at the **design-point (a)** under prescribed crossed fields  $E \times B$ , where  $E$  is the imposed electric field and  $B$  is the imposed magnetic field; this stage checks charged-particle confinement behavior, not fusion gain (here  $Q = P_{\text{fusion}} / P_{\text{input}}$ ). A separate operations simulator integrates the test stand for pad procedures.

In this design, a fuselage-integrated dorsal S-duct scoop feeds a single-spool **compressor–turbine** train with **externally heated Brayton** air (no combustion in the core path).

Under **(a)** the simulated plant reaches the **3.5 MW** headline while satisfying level-1 gates (OD plant + U1–U4; see **Benchmark Methodology**). Under **(b)** expect a large shortfall — that is the quantitative “mountain.” Under **(c)** the inverse states what effective reactivity and knobs would be required on literature  $\sigma v$ .

## Phases

The Orbitron program uses **Phase** to mean a **hardware and operations milestone** on the test stand — where the rig runs, what reaction mass is used, and how fusion heat is offloaded. A phase is **not** a single simulation timestep, a proof-chain step number, or a frame in an assembly animation.

### Phase 1 — Benchtop (stationary ultra-high vacuum)

Phase 1 is the **stationary laboratory** configuration. The operator sequence is: evacuate the chamber and confirm vacuum interlocks; align and arm the UV laser on **solid boron-11** targets; enable high-voltage bias on the central cathode through safety interlocks; introduce **hydrogen** while watching beam and fusion diagnostics. Fuel enters as a laser ablation plume plus controlled  $H_2$  flow. Ions are confined and accelerated in the electrostatic trap (with weak axial magnetic field for  $E \times B$  electron neutralization); the laser delivers fuel, it does not steer the beam.

Experiments and reports scoped to **Phase 1** assume this bench layout: core geometry and pad interlocks appropriate to a sealed UHV vessel, without wind-tunnel airflow or a spooled compressor–turbine train.

### Phase 2 — Wind-tunnel rig (ground Brayton)

Phase 2 adds a **ground air-breathing path**: simulated or blower-fed intake, compressor light-off, bleed through inlet guide vanes into the containment jacket annulus, and — once flow is stable — the same Phase 1 vacuum, laser, and high-voltage interlocks inside the running duct. Fusion-heated mixed gas drives a **turbine** that sustains the **compressor**;

exhaust leaves through silencing ducting. The primary reaction mass is **ingested air**, not tanked propellant. Cryogenic wall thermal services (for example methane cooling of the first wall and HTS magnet) belong to the integrated plant; they are support systems, not the Phase 1 fusion fuel path.

## Relation to the rest of this report

Sections on the test stand, thermal zoning, and jet closure mix **Phase 1 core physics** with **Phase 2 propulsion plumbing** where the narrative needs both. The numbered **proof-chain steps** in the forward model (geometry through plant and validation) are a separate software validation ladder; they support both phases but do not map one-to-one to each operator action in Phase 1 or Phase 2.

## Benchmark Methodology

### Purpose

Reproducible **integration + physics-envelope** benchmark: CAD layout, WarpX PIC electron loading, 0D plant closure with U1–U4 gates, and unobtainium inverse. **Not** a license to operate a reactor or a claim that p-<sup>11</sup>B Orbitron fusion is demonstrated at 3.5 MW.

### Validation levels

Claims in this report are tagged by **level** — what each part of the toolchain is allowed to prove:

- **0** — Pad interlock sequence — Correct startup order before fueling and reaction
- **1** — 0D plant + U1–U4 gates — **3.5 MW** headline, jet closure, materials limits (per  $\sigma v$  branch)
- **2** — WarpX PIC (electron ring) — Density and  $E \times B$  loading at design voltage — **not** fusion gain
- **3** — p-<sup>11</sup>B channel + burn models —  $\langle \sigma v \rangle (T_i) \times \text{fueling} \times \text{volume}$ ; laminar / clump checks
- **4** — *Future* — Transport-integrated reactivity without analytical surrogate blend

Never write “WarpX proves 3.5 MW.” Level **2** validates electrons in prescribed fields only; level **1** is plant accounting at the chosen reactivity branch.

**HTS** — **high-temperature superconductor** bore magnet (U3). **REBCO** — **rare-earth barium copper oxide** tape; **YBCO** (yttrium barium copper oxide) is the common member of that family. Both are used below as published anchors for achievable field in compact solenoids.

### Three scenarios (only these)

- **pretend** — (a) Design target — design (calibrated) — 600 kV, 2 T, unity U1–U4
- **today** — (b) COTS + experiment — literature — Avalanche-class **300 kV**; same pad fueling as (a); wall/HTS at published limits
- **minimum** — (c) Inverse minimum — literature — Constrained stress inverse (literature  $\sigma v$ )

No aspirational “5-year SOTA” forward row. R&D narrative lives in gap analysis, not a second pretend run.

### (a) Pretend

- Primary **forward model** (steps 0–8) runs here.
- **Design validated** at level 1 means **calibrated plant closure**, not measured fusion yield.
- WarpX step 01 figures are labeled **design-point (a)** (level 2).

### (b) Today

- **Fusion**: literature  $\langle\sigma v\rangle$  only (literature peak in the p-<sup>11</sup>B reactivity model).
- **Pad fueling**: same H<sub>2</sub> flow, laser rate, and throttle as (a) — shortfall is physics/materials, not retuned fuel.
- **Voltage**: Avalanche Orbitron public milestone **300 kV sustained** (D<sub>2</sub> hardware; topology anchor for Orbitron-class, not p-<sup>11</sup>B yield).
- **U2 wall**: ~1.0 MW/m<sup>2</sup> steady limit class (tokamak/DEMO PFC scoping).
- **U3 HTS**: 0.8 T effective bore vs 2 T design → HTS capability scale 0.4 (REBCO ~0.8 T @ 77 K class demos).

### (c) Minimum

- **Constrained stress inverse** on literature  $\sigma v$ : minimize fusion reactivity scale  $\eta_{\text{react}}$  subject to gross power  $\geq$  target and U1–U4 inequalities (Levenberg–Marquardt / trust-region least squares on constraint residuals).
- **success = true** only if design validated and no hard spec FAIL — otherwise **(c) is infeasible**.
- **Margin inverse**: design  $\sigma v$ ; minimize knob distance from nominal under the same gates — should approximate **(a)**.

### Inverse solver rules

- **Stress (c)**: find the **minimum**  $\eta_{\text{react}}$  (on literature  $\sigma v$ ) for which a feasible point exists; do not treat power-only fits as solutions.
- **Margin (a check)**: design  $\sigma v$ ; prefer knobs near 1.0×.
- **Forward confirmation**: design  $\sigma v$  at margin-inverse knobs — internal consistency only.

### Report outputs

1. **Scenario comparison table** — (a)(b)(c): gross power,  $\sigma v$  branch ratio, effective gap, level-1 gates pass.
2. **Stress section** —  $\eta_{\text{react}}$  required, design/literature  $\langle\sigma v\rangle$  ratio at operating ion temperature.
3. **PIC** — design-point (a) unless a second run is added later.

**Gate before publishing**: (b) gross power must be **much** lower than (a). If not, (b) anchors are wrong.

### Radial thermal zoning (resolves HTS vs hot air)

p-<sup>11</sup>B is **aneutronic** — the first wall catches **alphas**, **bremsstrahlung X-rays**, and **charge-exchange** losses, not a neutron blanket. The **HTS magnet is not the air “jacket.”** Compressed air must **not** flow over the

cryogenic coil; it flows in an **annulus inside the magnet radius**, over a **hot first-wall sheath**.

**Inside-out stack** (Phase-1 radii at  $r_{\text{anode}} = 4$  cm):

- Cathode — 0 → 1 cm — -600 kV wire
- Plasma vacuum — 1 → 4 cm —  $E \times B$  + beams only
- **First wall** — 4 cm — Absorbs  $\alpha$  / X-ray / CX; **800–1000 °C** class
- **Air annulus** — 4 → 6 cm — Brayton working fluid **heated** by hot wall
- **Cryostat (vacuum + MLI)** — 6 → 7.5 cm — Blocks heat leak to magnet
- **HTS solenoid** — 7.5 → 10 cm — 2 T through vacuum; **CH<sub>4</sub>** removes parasitic leak only
- Casing — > 10 cm — Structure, services

**Two thermal fluids, two jobs:**

- U2 intercept — Liquid **CH<sub>4</sub>** — Remove high-grade wall load so the sheath survives
- Brayton — Compressed **air** — Enthalpy for turbine / nozzle — from **hot wall + ash mixer**, not from quenching HTS
- U3 cryostat — Liquid **CH<sub>4</sub>** (closed) — Hold magnet at ~113 K; separate from air temperature

**0D energy split** (level 1):  $\text{first\_wall\_kw} \rightarrow \text{CH}_4 \text{ intercept} + \text{air annulus}$ ;  $\text{brayton\_thermal\_kw}$  drives the jet surrogate;  $\text{gross\_mw}$  remains the fusion headline. Passing U1–U4 does **not** yet size cryostat mass, refrigeration electrical power, or a dedicated **liquid-to-air heat exchanger** if the annulus alone cannot reach  $T_3$ .

### Non-goals

- No secondary “judgment” scenario row beyond (a)(b)(c).
- Gap narrative does not override solver numbers.
- No claim that air washes the HTS pack or that the magnet absorbs  $\alpha$  / X-ray directly.

### The air-breathing Brayton cycle

The propulsion thermodynamic cycle is an **open-loop, externally heated Brayton cycle** on a **single compressor–turbine spool**. Conventional jets add heat by burning fuel in the working fluid [1, 3]; here enthalpy is added when compressed air passes through a **hot annulus around the fusion first wall** (and mixes with a small **<sup>4</sup>He ash** stream) — **not** by flowing over the cryogenic HTS magnet. The working fluid remains atmospheric air; there is no combustion chemistry in the main path.

**Pad start:** an electric starter (APU cart) drives the shaft; **bleed air** opens the bellmouth → compressor annulus so the machine can ingest and compress air before fusion is armed. **Cruise / light-off:** after the hot section raises gas enthalpy, a **turbine** expands part of the flow to supply compressor shaft work; the pad motor disengages. **Nozzle expansion** of the core stream produces thrust.

### Cycle Configuration and the Dorsal Intake

A dorsal S-duct scoop (727-style fuselage intake) captures ambient air. With bleed open, a fraction of ingested air may be routed through

a **compressor bleed path** (parasitic flow that does not pass through the fusion jacket) while the **core path** supplies the heated stream to the turbine and nozzle. Bleed establishes minimum compressor flow during ground start and reduces the effective core mass flow available for thrust.

## The Four Stages of the Cycle

In air-standard notation:

1. **Isentropic Compression (1 → 2):** Air enters the S-duct and is compressed in the rotary compressor. **Before turbine takeover:** shaft work is supplied by the **pad electric starter** ( $W_{\text{shaft,in}} = W_{\text{c,elec}}$ ). **After takeover:** shaft work is supplied by the **turbine** ( $W_{\text{shaft,in}} = W_t$ ). Compression raises P and T.
2. **Isobaric Heat Addition (2 → 3):** Core-path air passes through the **first-wall annulus** (hot metal at ~800–1000 °C class) at nearly constant pressure. No fuel is mixed into the air stream; the HTS magnet sits **outside** a vacuum cryostat and is not the heat-exchanger surface.
3. **Split Expansion (3 → 4<sub>t</sub>, 4<sub>t</sub> → 4):** Hot gas first expands through the **turbine** (3 → 4<sub>t</sub>) to deliver wt per unit mass to the shaft. The remaining enthalpy expands through the **nozzle** (4<sub>t</sub> → 4) to exhaust velocity and thrust. Before takeover, the starter still provides wc and only the nozzle branch carries net propulsive acceleration for the core stream.
4. **Isobaric Heat Rejection (4 → 1):** Exhaust and bleed dump to the atmosphere; fresh air enters the scoop [1, 2].

## Bleed Air and Effective Mass Flow

Let  $\dot{m}_{\text{in}}$  be total corrected inlet mass flow (compressor command × spool capability), and  $\beta$  the **bleed mass fraction** when the bleed valve is open ( $0 \leq \beta < 1$ ):

$$\dot{m}_{\text{bleed}} = \beta \dot{m}_{\text{in}}, \dot{m}_{\text{core}} = (1 - \beta) \dot{m}_{\text{in}}$$

Only  $\dot{m}_{\text{core}}$  passes through the fusion jacket, turbine, and nozzle thrust bookkeeping. Compressor shaft work is evaluated on  $\dot{m}_{\text{in}}$  (the machine pumps the bleed stream as well).

Pad / surrogate effective compressor command (steps 03–07):

$$c_{\text{eff}} = c \cdot 1_{\text{bleed}} \cdot s_{\text{spool}}$$

with  $s_{\text{spool}} = 0.12$  (bleed only), 0.42 (starter engaged), or 1.0 (fusion armed and starter off — **turbine takeover**).

## Mathematical Efficiency and Power Balance

Ideal-cycle thermal efficiency vs pressure ratio  $r_p = P_2/P_1$  and  $\gamma \approx 1.4$  [1]:

$$\eta_{\text{th}} = 1 - (1)/(r_p^{\gamma} - 1)^{1/\gamma}$$

**Compressor shaft work** per unit mass (on total inlet flow):

$$w_c = (C_p (T_2 - T_1))/(\eta_c)$$

**Pad start (electric drive):**  $W_{\text{c,elec}} = \dot{m}_{\text{in}} w_c / \eta_{\text{motor}}$ .

**Turbine takeover:** turbine specific work wt must balance wc (mechanical efficiency  $\eta_{\text{mech}}$  on the spool):

$$wt \approx wc / \eta_{\text{mech}}, Wt = \dot{m}_{\text{core}} wt$$

**External heating** on the core path:

$$q_{\text{in}} = C_p (T_3 - T_2), Q_{\text{in}} = \dot{m}_{\text{core}} q_{\text{in}}$$

**Nozzle / jet power** (core stream):

$$w_j = \eta_n C_p (T_{4_t} - T_4), P_{\text{jet}} \approx 1/2 \dot{m}_{\text{core}} v_e^2$$

**Closure check (step 07):** with thrust F and  $\dot{m}_{\text{core}}$  used in  $F^2/(2\dot{m}) \approx P_{\text{jet}}$ .

At takeover, net propulsive power must exceed pad electrical draw; in steady cruise the compressor is sustained by Wt from fusion-heated gas, not ship battery.

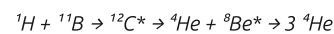
## Engineering Notes

- **Bleed** is required for ground start and is modeled as a reduction in  $\dot{m}_{\text{core}}$ ; closing bleed interlocks fusion ignite in the pad sequence.
- **Turbine takeover** is procedural in FlightGear (starter off after light-off); the OD plant uses  $\text{sspool}=1$  only when armed and starter is disengaged.
- **Externally heated** operation avoids combustion product chemistry in the core path [1, 8] while retaining a conventional shaft-coupled turbomachine [3, 9].

## Why p-<sup>11</sup>B fusion?

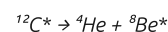
### The p-<sup>11</sup>B multi-step fusion reaction

The proton-boron (p-<sup>11</sup>B) fusion reaction is a sequential, multi-step nuclear process that transitions through short-lived intermediate states rather than occurring simultaneously. The primary reaction pathway is written as:

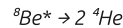


The step-by-step sequence is characterized by the following phases:

- **Step 1: Proton Capture and Compound Nucleus Formation:** A proton (<sup>1</sup>H) fuses with a boron-11 (<sup>11</sup>B) nucleus, creating a highly unstable, excited compound carbon-12 (<sup>12</sup>C\*) state. When formed via the dominant 675 keV resonance, the carbon-12 exists primarily in a 1<sup>-</sup> quantum spin state.
- **Step 2: Primary Alpha Emission ( $\alpha_0$ ):** In approximately 10<sup>-21</sup> seconds, the excited <sup>12</sup>C\* nucleus decays by ejecting a primary helium-4 nucleus (alpha particle,  $\alpha_0$ ), leaving behind an excited beryllium-8 (<sup>8</sup>Be\*) nucleus:



- **Step 3: Beryllium Splitting ( $\alpha_1, \alpha_2$ ):** The resulting <sup>8</sup>Be\* nucleus is highly unstable and splits within a fraction of a femtosecond into two secondary helium-4 nuclei ( $\alpha_1$  and  $\alpha_2$ ):



This step-by-step mechanism explains why aneutronic reactor designs yield safely contained charged particles instead of high-energy neutrons.

## Thermodynamic Threshold and the Coulomb Barrier

Initiating this reaction requires overcoming the electrostatic repulsion between the positive charges of the proton ( $Z1 = 1$ ) and the boron-11 nucleus ( $Z2 = 5$ ).

To calculate the classical Coulomb barrier ( $V_c$ )—the energy needed for the proton to touch the surface of the boron-11 nucleus—we use:

$$V_c = (1)/(4\pi\epsilon_0) (q1 q2)/(R1 + R2)$$

With standard nuclear radii formulas ( $R \approx 1.2 \times A^{1/3}$  fm): *Proton radius* ( $R1 \approx 1.2$  fm) Boron-11 radius ( $R2 \approx 1.2 \times 11^{1/3} \approx 2.67$  fm) \* Total separation distance ( $R = R1 + R2 \approx 3.87$  fm)

Using the electrostatic constant  $k_e = (e^2)/(4\pi\epsilon_0) \approx 1.44$  MeV·fm:

$$V_c \approx (1.44 \times 1 \times 5)/(3.87 \text{ fm}) \approx 1.86 \text{ MeV}$$

According to classical mechanics, overcoming this barrier would require a kinetic energy of 1.86 MeV (equivalent to over 21 billion Kelvin).

However, quantum-mechanical tunneling allows the proton to penetrate the barrier at lower energies. The probability of tunneling (the reaction cross-section,  $\sigma$ ) is highly dependent on the energy of the colliding particles and is dominated by two low-energy resonances in the center-of-mass frame: 1. **The 148 keV Resonance:** A narrow peak where the cross-section reaches  $\approx 0.1$  barns. 2. **The 675 keV Resonance:** A much broader and higher peak where the cross-section reaches its absolute maximum of  $\approx 1.4$  barns.

Because the 675 keV resonance is over ten times stronger and significantly broader, the absolute maximum rate of fusion occurs when the bulk of the colliding particles possess energies near this peak.

Using the Boltzmann relation, temperature ( $T$ ) and average kinetic energy ( $E$ ) are linked by:

$$E = k_B T$$

Given the conversion factor  $1 \text{ eV} \approx 11,604 \text{ K}$ , we can find the thermal equivalent of this peak energy:

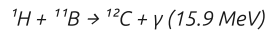
$$E = (8,000,000,000 \text{ K})/(11,604 \text{ K/eV}) \approx 689,400 \text{ eV} \approx 690 \text{ keV}$$

Because 675 keV translates directly to  $\approx 7.8$  billion Kelvin, a thermal plasma must be maintained at approximately **8 billion Kelvin** to align the average thermal motion of the ions with the optimal resonance peak.

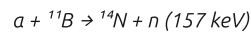
## Side-Channel Reactions

In a real operating reactor, minor side-channel reactions occur alongside the primary pathway, producing penetrating, unconfined "extra"

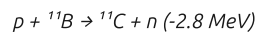
1. **Radiative Capture (occurrence rate  $\approx 0.01\%$ ):** Occurs when the compound carbon-12 nucleus drops directly to its ground state instead of splitting, emitting a high-energy gamma-ray photon:



1. **Alpha-Boron Interactions (occurrence rate  $\approx 0.1\%$ ):** Fast alpha particles produced in the primary reaction collide with the boron-11 fuel, generating a minor neutron flux:



1. **High-Energy Endothermic Reaction:** Protons in the high-energy tail of the distribution trigger a reaction that produces an endothermic neutron and radioactive Carbon-11:



### Energy output of the p-11B fusion reaction

The primary p-<sup>11</sup>B reaction yields a total net energy of 8.7 MeV per reaction. This net energy is converted entirely into the kinetic energy of the three resulting helium-4 (alpha) particles:

- The primary alpha particle ( $\alpha_0$ ) carries approximately 3.76 MeV.
- The two secondary alpha particles ( $\alpha_1$  and  $\alpha_2$ ) split the remaining energy, carrying approximately 2.46 MeV each.

Because the energy is released as the kinetic energy of charged alpha particles rather than high-energy neutrons, the reaction is primarily aneutronic. These charged particles can theoretically be contained electromagnetically or converted directly into electrical energy, reducing reliance on conventional thermal cycles.

However, minor energy losses and alternative emissions occur via the side-channel reactions noted above, which release energy in the form of a 15.9 MeV gamma ray, a 157 keV neutron, or an endothermic loss of 2.8 MeV (accompanied by a neutron and Carbon-11).

### Can a practical aircraft operate in the presence of an 8GK reaction?

To evaluate whether a practical aircraft can operate in the presence of an 8 billion Kelvin (8 GK) reaction, we must distinguish between thermal and non-thermal (beam-like) plasma systems, and then assess the real-world thermal and radiation loads.

### Thermal Heat vs. Directed Kinetic Energy (The Orbitron)

In a thermalized plasma system (such as a tokamak), particles collide constantly, randomizing their velocities into a Maxwell-Boltzmann distribution. In this configuration, an 8 GK temperature represents chaotic, high-density thermodynamic heat. If this plasma makes direct contact with a physical wall, it will instantly vaporize it.

An Orbitron-style reactor, however, is a **non-thermal system**. By utilizing a 600 kV central cathode, positive ions (protons and boron-11) are

accelerated to a kinetic energy of 600 keV. While 600 keV mathematically equates to ~7 to 8 billion Kelvin, the ions are locked in highly organized, directed orbits (which can be further organized using laminar flow techniques).

Because the velocity is directed rather than chaotic, this is directed kinetic energy rather than ambient thermal heat. In an ideal scenario with absolute vacuum and perfect confinement, the containment vessel walls would remain at room temperature.

## Reality of Losses and Airborne Engineering Constraints

In a practical engineering design, confinement is imperfect, and three primary physical mechanisms transfer energy to the containment vessel, creating high-temperature thermal and radiation loads:

1. **Bremsstrahlung (X-Ray) Radiation:** When energetic 600 keV electrons and boron ions ( $Z=5$ ) interact, the electrons deflect and decelerate rapidly, releasing their kinetic energy as high-energy X-ray photons. Because these radiative losses scale with  $Z^2$ , the Bremsstrahlung losses in a p- $^{11}\text{B}$  reactor are 25 times higher than in a hydrogen-hydrogen system. These X-rays pass directly through electrostatic and magnetic confinement fields, presenting a continuous, high-intensity radiant energy flux that will rapidly heat the containment walls to their melting point without active cooling.
2. **Charge-Exchange (CX) Losses:** Because an absolute vacuum is practically impossible to maintain, background neutral gas molecules will always be present. A fast, orbiting 600 keV ion can capture an electron from a cold neutral atom, instantly becoming a fast neutral. No longer bound by the cathode's electrostatic field or magnetic confinement, this fast neutral flies out in a straight tangent line, slamming into the containment wall and converting its kinetic energy into localized thermal heat and material sputtering.
3. **Scattering and Grid/Cathode Collisions:** Ion-ion collisions cause scattering. Some scattered particles spiral outward to strike the containment walls, while others lose angular momentum and collide directly with the 600 kV central cathode, risking ablation or melting of the electrode.

## Implications for Aircraft Integration

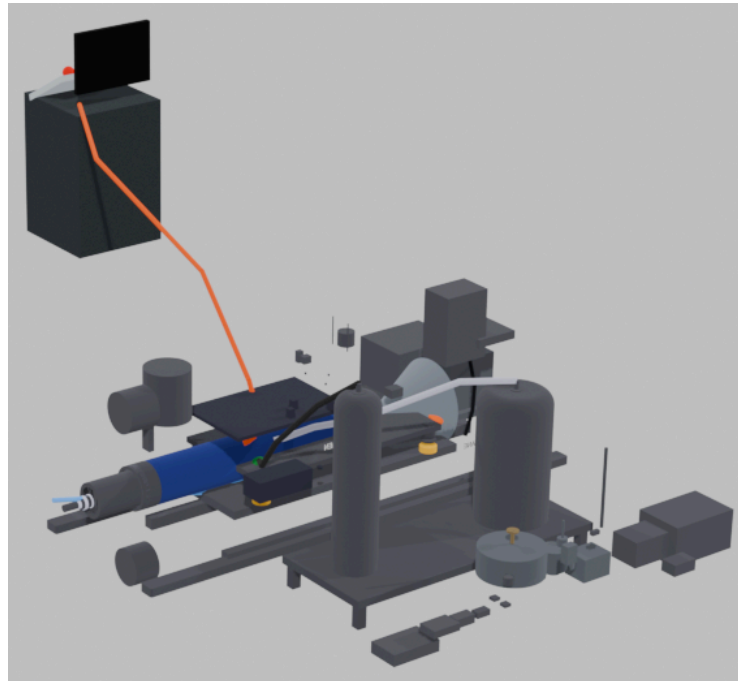
For an aircraft to operate with a p- $^{11}\text{B}$  reactor, the system must handle these high-temperature thermal loads and radiation fluxes. This introduces several severe engineering constraints:

- **Active Cooling Systems:** Heavy shielding or active cooling (such as water-cooled copper jackets) is required to manage the intense Bremsstrahlung X-ray flux and prevent the containment vessel from melting.
- **Vacuum Maintenance:** High-performance, robust vacuum systems are necessary to minimize charge-exchange losses and prevent localized wall damage.
- **Radiation Shielding:** Effective shielding is required to protect the aircraft's crew, avionics, and airframe from continuous X-rays, as well as the minor neutron and gamma fluxes produced by side-channel reactions.
- **Weight Constraints:** The weight of the active cooling systems, radiation shielding, and high-voltage power supplies must be balanced against the lift requirements of a practical aircraft.

While the non-thermal nature of an Orbitron prevents the reactor walls from melting via direct plasma contact, the secondary heat and radiation loads from Bremsstrahlung, charge-exchange, and particle scattering present significant engineering challenges for integration into a practical aircraft.

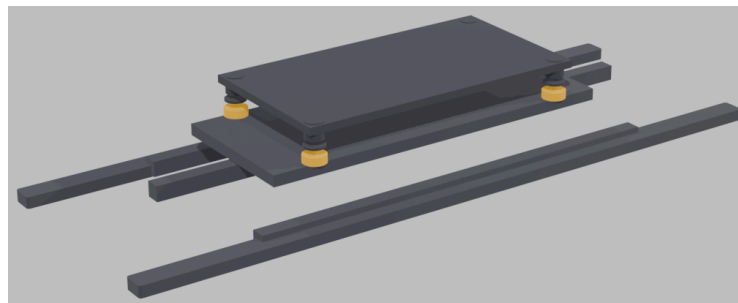
### The Phase-1 test stand

Propulsion runs  $-X \rightarrow +X$  from bellmouth intake to nozzle exit. Cryogenic  $H_2$  and  $CH_4$  services sit on the pad deck; the electrostatic core, laser ablation line, and Phase-2 Brayton hardware share one logical layout. Labels below (**LAB-01**, **CORE-01**, **AIR-01**, ...) are engineering tags for cross-reference only.



LAB-01

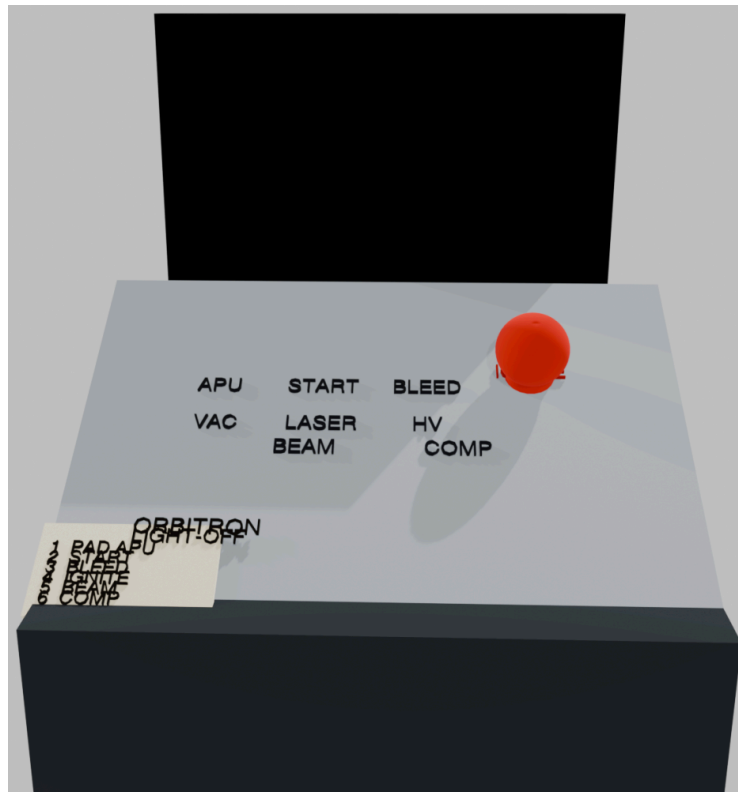
### TS-01 — Thrust sled & load cells



TS-01

Four corner load cells integrate thrust and moment for jet-shaft power bookkeeping. The engine mount frame sets the pivot height for the Brayton train relative to the pad deck.

### CTRL-01 — Control panel & pad interlocks



CTRL-01

Operator console for pad power, compressor bleed, vacuum interlock, laser arm, and high-voltage enable before ignition. The interlock chain mirrors a real hot-fire sequence; fueling and reaction physics stay off until the pad is in a safe steady-ready state.

### CORE-01 — Electrostatic Orbitron core (Phase 1)

CORE-01

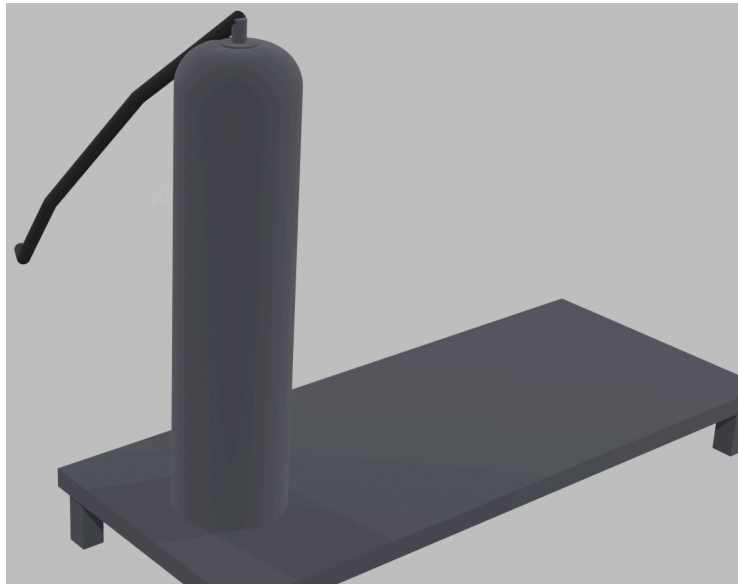
**Radial stack (inside → outside)** — normative layout for p-<sup>11</sup>B; see **Benchmark Methodology — Radial thermal zoning** and level-1 thermal split in this report.

1. **Cathode wire** (on-axis, ~-600 kV) and **plasma vacuum bore** — keV H<sup>+</sup> / B<sup>+</sup> from tangential NBI and solid <sup>11</sup>B laser ablation; no Brayton air in the bore. Energy leaves as <sup>4</sup>He alphas, bremsstrahlung X-rays, and charge-exchange — not a significant neutron flux.

2. **First wall / anode sheath** (Outer\_Anode\_Grid) at  $r \approx 4$  cm — the **heat catcher**: absorbs  $\alpha$  and X-ray / CX load (~400 kW class in the 0D model). Runs **hot** (~800–1000 °C class) on the plasma side.
3. **Air annulus** — compressed air flows **between the hot first wall and the cryostat**, **inside** the magnet bore radius. It **heats** the air for the Brayton train; it does **not** wash or cool the HTS winding.
4. **Cryostat** — vacuum gap + **MLI** (target ~1.5 cm radial in the zoning budget). Blocks conduction and convection from the hot air channel to the coil.
5. **HTS solenoid** (Magnet, ~7.5–10 cm outer radius) — **outside** the cryostat; **B**  $\approx 2$  T penetrates the bore through vacuum. **Liquid CH<sub>4</sub>** (~113 K) removes **parasitic cryo leak only** — not the megawatt first-wall stream.

**CAD:** glTF exports **five radial zones** as separate annular solids — Central\_Cathode\_Wire, Outer\_Anode\_Grid (first wall), Air\_Annulus\_Channel, Cryostat\_Vacuum\_Gap, Magnet (HTS), plus Reactor\_Bay\_Inlet\_Shroud on the engine train. Radii match assembly.radial\_thermal\_stack / Phase-1 benchmark (4–10 cm OD stack).

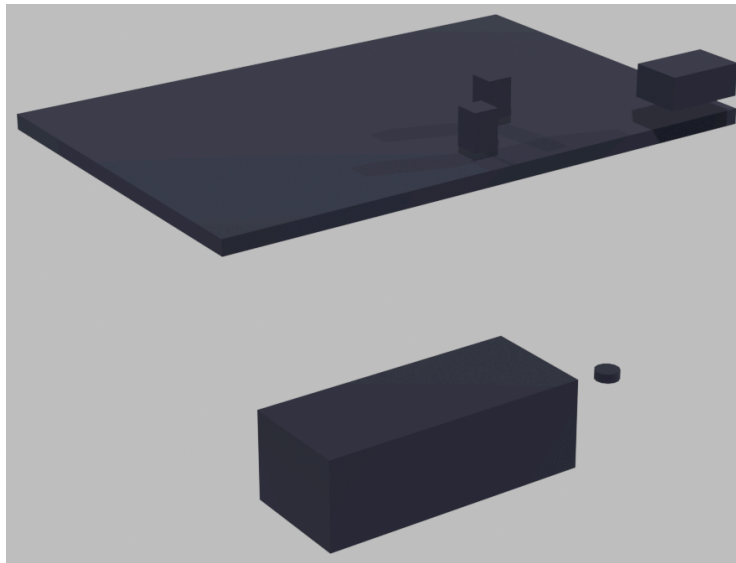
### INJ-H2-01 — Hydrogen proton feed



INJ-H2-01

Cryogenic **H<sub>2</sub>** tank and trunk line feed the neutral-beam path for proton inventory. The benchmark fueling point targets strong mixing with the **<sup>11</sup>B** ablation line near an **8:1 H<sub>2</sub>:laser** duty ratio.

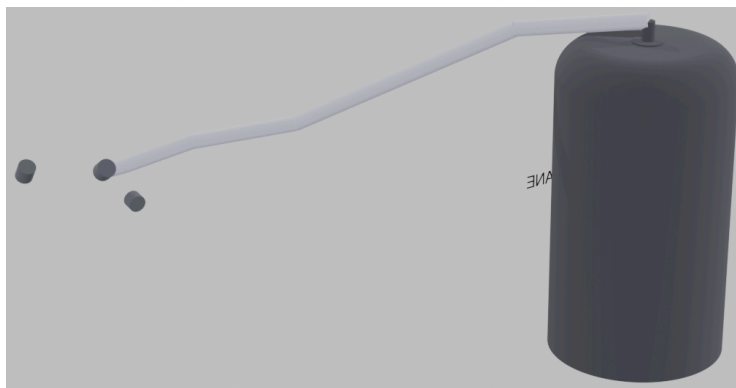
### INJ-B11-01 — Solid <sup>11</sup>B laser ablation



INJ-B11-01

Q-switched **Nd:YAG** at **355 nm** cold-ablates solid **<sup>11</sup>B** disks in the vacuum chamber viewport line. Repetition rate sets boron delivery into the electrostatic well alongside the hydrogen feed.

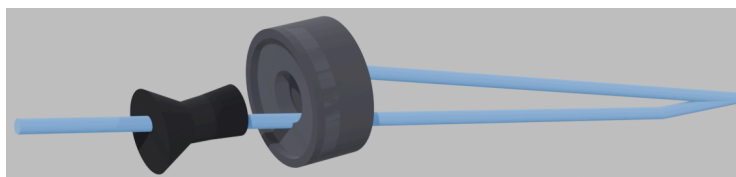
#### U2-CH4-01 — CH<sub>4</sub> wall-thermal loop (Unobtainium U2)



U2-CH4-01

Two **CH<sub>4</sub>** duties, not one: (**U2**) internal channels on the **first wall** remove the high-grade  $\alpha$  / X-ray / CX load (~55% of first\_wall\_kw in the 0D split); (**U3**) a closed cryostat loop on the **HTS solenoid** removes **parasitic leak** through the vacuum gap. **CH<sub>4</sub> does not cool the Brayton air stream.**

#### AIR-01 — Air-breathing Brayton train (–X intake → +X nozzle)



AIR-01

Bellmouth and S-duct intake on **–X** feed a co-axial **compressor–turbine** spool: pad **bleed** opens the path, an **electric starter** cranks the shaft, then after fusion light-off the **turbine** drives the compressor. Core-path

air is heated in the **annulus around the hot first wall**

(Reactor\_Bay\_Inlet\_Shroud region) — **not** by flowing over the cryogenic HTS pack — then expands through the **+X** nozzle for thrust on the sled.

## Governing equations

State evolution for the forward model (stages 0–8). Each stage defines a state vector, initial condition, and discrete update.

Each step defines a **state vector**  $S$ , an **initial condition**  $S(0)$ , and a **discrete update**  $S(tk+1) = f_k(S(tk))$ . Subscripts name the step; time index  $k$  is step-specific (PIC step, channel frame, or algebraic).

Symbols are factored so **fuel** and **Brayton compressor** do not appear in step 01 WarpX.

**Shared geometry (from step 00):** bore radii  $r_c$ ,  $r_a$ , length  $L$ , cathode voltage  $V_c$ , axial field  $B$ .

**Pad levers (naming):**

- $\tau$  — Ring density scale ( $W/S$ ) — Yes — initial  $n_e$  — 01
- $p$  — Cathode pulse / shear ( $I/K$ ) — Yes —  $n_e$  at  $t=0$  and  $E(t)$  ramp — 01
- $c$  — Compressor ( $U/J$ ) — **No** — 06 plant ( $\dot{m}$ )
- $H_2$  sccm, laser Hz — Step 00 injectants — **No** — 03–04 fueling

## Step 0 — Design compile

**State**  $S_0 = \{G, I, Y\}$ :

- $G = (r_a, r_c, L, V_c, B)$  geometry
- $I = (\dot{Q}_{H_2}, \text{flaser})$  injectants (sccm, Hz)
- $Y$  = design specification bundle

**Initial:** operator sets geometry and pad interlocks.

**Update (algebraic, one shot):**

$$S_1 = \text{Compile}(Y, G, I)$$

No  $f(S(t))$  — not a time integrator.

## Step 1 — Electron-ring WarpX (level 2)

**State** at PIC index  $k$ :

$$S_1(k) = \{(x_i, v_i, w_i)\}_{i=1}^{Np}, tk, a_k$$

Species: **electrons only**. Prescribed fields (not from plasma Poisson):

**Initial**  $S_1(0)$  before first push:

- Macroparticle set  $\{(x_i, v_i, w_i)\}$  for species **electrons** only.
- Simulation time  $tT = T \Delta t$ .

- Prescribed fields (not solved from plasma):
- $E0(x,z)$  = analytic cylindrical cathode–anode radial field from step 00  
 $V_c, r_c, r_a$ .
- $B = (0, B, 0)$  constant from step 00.
- Initial density scale (set once at  $t=0$ , before first push):

$$n_e = n_{e,base} (0.15 + 0.85 \tau) (0.65 + 0.35 p)$$

- Cathode ramp (every step, multiplies the electric field E only):

$$a(t; p) = 0.30 + (0.88 + 0.12 p - 0.30) t/tend \quad t \leq tend \quad 0.88 + 0.12 p \quad t > tend$$

with  $tend = 0.35 Nsteps \Delta t$ .

**Not in state:** fuel species, compressor, arc seed, inject beams, Poisson solve.

**Update**  $k \rightarrow k+1$  (F1):

$$E(x,z,tT) = E0(x,z) a(tT; p), \quad B = const$$

$$a_i = (qe)/(me) (E(x_i,tT) + v_i \times B)$$

$$v_i T+1 = v_i T + \Delta t a_i, \quad x_i T+1 = x_i T + \Delta t v_i T+1$$

$$\rho_{_e T+1}(x,z) = Deposit(\{x_i T+1, w_i\})$$

$\tau$  and  $p$  affect  $n_e$  only at initialization;  $p$  also enters  $a(t)$  each step.  $c$  does not appear.

## Step 2 — PIC reduce

**State:**  $S2 = (p_{_e sim}, nedesign)$  from stage 1 output and control levers.

**Initial:** last electron-density diagnostic from stage 1.

**Update** (algebraic):

$$\rho_{_e, p95} = P95(\rho_{_e} \text{ in annulus } [0.9 r_c, 0.95 r_a])$$

$$\rho_{_e, norm} = clamp_{0.05, 3}(\rho_{_e, p95}/n_e)$$

## Step 3 — Fusion channel (s–r)

**State** at frame  $k$ :

$$S3(k) = (nk(s,r), Ck, tk)$$

- $nk$  = fusion-relevant density on  $(s,r)$  grid inside bore  $r \leq r_a$
- $Ck$  = clump index =  $P95(n) / median(n)$  over active cells
- Controls (held fixed per run):  $\dot{Q}_{H2}$ , flaser,  $\rho_{_e, norm}$ , laminar flag  $L$ , compressor effectiveness  $ceff$

**Injection rate scale** (injection lever  $\lambda$ ):

$$\lambda = \dot{Q}_{H2} / \dot{Q}_{H2,ref} \sqrt{f_{laser} / f_{laser,ref}} \quad (ref = 80 \text{ sccm}, 10 \text{ Hz})$$

**Axial stir** (not fuel mass — Brayton path proxy):

$$ceff = c \cdot sspool, \quad sspool = 0 \text{ bleed closed } 0.12 \text{ bleed only } 0.42 \text{ starter on (electric shaft) } 1.0 \text{ armed, starter off (turbine takeover)}$$

Bleed splits inlet flow:  $\dot{m}_{bleed} = \beta \dot{m}_{in}$ ,  $\dot{m}_{core} = (1-\beta)\dot{m}_{in}$  (core path through jacket / turbine / nozzle).

**Initial** S3(0):

$$n0(s,r) = nseed \cdot \lambda$$

If laminar **OFF**: multiply by deterministic ripple +  $\mathcal{N}(0,1)$  noise (seed from stochastic seed).

1. **End inject** ( $H^+$  /  $B^+$  blobs at  $s \approx \text{ends}$ ):  $n \pm \Delta t \cdot \text{Ainj}(\lambda, \text{mix}, \tau, p) \cdot \text{Gaussians}, r$
2. **Clump seed** (only if  $L=0$  and  $k > K/8$ ): mid-bore Gaussian blobs  $\propto \lambda$
3. **Radial diffusion**:  $n \pm \Delta t \cdot \text{Deff}(L, p, \tau, B, \text{mix}) \cdot \partial^2 n / \partial r^2$
4. **Axial advection**:  $n := \Delta t \cdot us(ceff, \tau) \cdot \partial n / \partial s$
5. **Stochastic noise** (only if  $L=0$ ):  $n \pm \sigma_{noise} \cdot nseed \cdot \mathcal{N}(0,1)$
6. **Clip**  $n \geq 10^9$ ; store  $Ck$ , reaction proxy  $R \propto np \cdot nB \cdot v$ .

**Post-run scalars:**

$$fuelcoupling\_norm = clamp_{0.2,3}(maxs, r \cdot nK / means, r \cdot n0)$$

## Step 4 — Fueling densities

**State:** S4 = (G, l,  $\rho_e$ , norm,  $\tau$ ,  $\rho$ ,  $\eta_{react}$ ).

**Initial:** step 00 injectants + step 02  $\rho_e$ , norm.

**Update** (algebraic f4):

$$V = n \cdot ra^2 \cdot L \cdot ffill, \quad np = sccmto\_n(\dot{Q}_{H2} \cdot \text{mix}, V, \tau_{res}), \quad nB = sccmto\_n(f_{laser} \cdot scaleB, V, \tau_{res})$$

$$Ti = Ti(Vc, \rho, \tau), \quad v = f_{pb11}(Ti)$$

$$\eta_{conf} = \eta_{react} \cdot g(\tau, \text{mix}) \cdot clamp(\rho_e, norm)$$

## Step 5 — p-11B burn power

**State:** S5 = ( $np$ ,  $nB$ ,  $Ti$ ,  $v$ ),  $\eta_{conf}$ ,  $V$ ).

**Initial:** step 04 output; nominal reactivity assumption  $\eta_{react} = 1$ .

**Update** (algebraic f5):

$$R = np \cdot nB \cdot v, \quad P_{fusion} = \eta_{conf} \cdot V \cdot R \cdot Erxn$$

$$shortfall = P_{target} - P_{fusion}$$

## Step 6 — 0D plant

**State:** S6 = (Pfusion, G, U, pad, fuelcoupling) with unobtainium U.

**Initial:** steps 03–05 + pad (including **compressor c**).

**Update** (algebraic steady solve f6):

$$\dot{m}_{in} = \dot{m}_0 \cdot c_{eff} \cdot h(\tau, \rho_{e,norm}, fuelcoupling), \dot{m}_{core} = (1-\beta) \dot{m}_{in}, \\ \dot{m}_{bleed} = \beta \dot{m}_{in}$$

$$P_{gross} = f_{plant}(P_{fusion}, \dot{m}_{core}, U)$$

Shaft: pad **electric starter** supplies Wc,elec until ignite + starter off;  
then **turbine** shaft work balances wc on the spool.

U1–U4 inequality checks (cathode |E|, wall, HTS, density).

## Step 7 — Jet closure

**State:** S7 = (F,  $\dot{m}_{core}$ , Pjet, Pgross,  $\eta$ ) from step 06.

**Update** (algebraic f7):

$$P_{from\_F} = (F^2)/(2\dot{m}_{core}), \epsilon_{closure} = |P_{from\_F} - P_{jet}|/P_{jet}$$

Gate:  $\epsilon_{closure} \leq 0.12$ .

## Step 8 — Validation export

**State:** full forward-model states S8 = {S0 ...S7}.

## Design point

Operating point for this benchmark — geometry, fueling, unobtainium knobs, and plant targets.

**Geometry & fields** — r anode m: 0.04; r cathode m: 0.01; length m: 1.2; V cathode v: 6e+05; B axial tesla: 2.

**Fueling** — h2 sccm: 135; laser ablation hz: 16.875; b11 target index: 0.

**Unobtainium knobs** — fusion reactivity scale: 1; field emission margin: 1; max wall heat flux W m2: 2e+06; ch4 cooling effectiveness: 1; hts capability scale: 1; beam coupling scale: 1.

**Plant targets** — target gross power mw: 3.5; jet propulsive efficiency: 0.55; heat kw at full: 4; beam screen kw per ma: 0.6; thrust lbf at full: 404; mass flow kgps at full: 84; density log10 at full: 11.

## Thermal architecture — this run

Radial zoning per **Benchmark Methodology**. Level-1 split of first-wall power:

- First-wall load — **400.0 kW**
- CH<sub>4</sub> wall intercept — **220.0 kW**
- Air annulus → Brayton — **180.0 kW**

- Ash mixer (model) — **71.5 kW**
- Brayton thermal total — **251.5 kW** (0.25 MW)
- HTS cryostat load — **0.22 kW**
- Cryostat radiative budget (est.) — **1.77 kW**
- Reactor OD (zoned) — **20.0 cm**
- Gross fusion headline — **3.575 MW**

Jet surrogate uses **Brayton thermal × propulsive efficiency**, not the full gross headline. Cryopant electrical power and HX area are **not** closed in this benchmark.

## Unobtainium design basis (U1–U4)

Closing **3.5 MW** requires simultaneous progress on emission, wall cooling, bore field, and p-<sup>11</sup>B reactivity — not independent tuning knobs.

- **U1 — Cathode emission:** 600 kV-class without vacuum arc — margin **1.0×**
- **U2 — First wall + CH<sub>4</sub> loop:** 2000000.0 W/m<sup>2</sup>, cooling **1.0×**
- **U3 — HTS bore** (high-temperature superconductor): 2 T at cryogenic temperature — scale **1.0×**
- **U4 — p-<sup>11</sup>B reactivity × beam coupling:** reactivity **1.0×**, coupling **1.0×**

(a) Pretend uses the **design-calibrated**  $\langle\sigma v\rangle$  curve. (b) Today uses **literature-class**  $\langle\sigma v\rangle$  (~10<sup>3</sup>× lower peak at operating T) plus experimental anchors in scenario\_anchors.yaml. (c) Minimum is the stress-inverse solve.

### U1 — Non-arcing cathode (–600 kV)

- **Geometry:** ~5 mm radius on-axis wire, ~5–10 cm radial gap to anode
- **Potential:** –600 kV DC (design), 400–600 kV operating band
- **Environment:** High vacuum bore, no sustained vacuum arc
- **Material (concept):** Tungsten–rhenium core with **defect-free / CNT-class** surface treatment
- **Performance:** Field emission suppressed so the well is electrostatic, not a lightning short to the wall

The simulator derives: allowable surface field (V/m), emission current density, and whether **600 kV** is consistent with gap geometry without arc — feeds U1 acceptance criteria.

### U2 — First wall (hot) + CH<sub>4</sub> intercept

- **Location:** First wall / anode sheath at plasma radius — catches **α**, **X-ray**, **CX** (not neutrons)
- **Wall load (design anchor):** ~400 kW steady boundary deposition (0D heat\_kw\_scale)
- **Wall material (concept):** Niobium C-103-class pressure boundary (or equivalent)
- **Face temperature:** 800–1000 °C class toward plasma; outer face feeds **air annulus**
- **Coolant:** Liquid methane ~–160 °C in internal channels — removes ~55% of wall load in 0D split

- **Limit:** Metal stays below failure; **CH<sub>4</sub>** must close flux limits independent of Brayton air

The simulator derives: CH<sub>4</sub> ṁ for the **intercept fraction** of first\_wall\_kw, not for cooling compressed air.

### U3 — HTS solenoid outside cryostat (cold zone)

- **Layout:** Outside vacuum cryostat; **B** penetrates bore; **no** direct α / X-ray flux on tape
- **Field:** 2.0 T axial **B** in the plasma bore (PIC / core specification)
- **Magnet (concept):** YBCO-class HTS tape solenoid (~7.5–10 cm outer radius in zoning budget)
- **Temperature:** ~113 K via liquid **CH<sub>4</sub>** cryostat loop (parasitic leak only — not the 400 kW wall stream)
- **Insulation:** Vacuum gap + MLI between **hot air annulus** and coil
- **Mass:** Light enough for SSTO vs copper — **cryostat + MLI mass not fully sized in OD**

The simulator derives: cryo load vs HTS capability scale; radiative leak budget vs air-channel temperature.

### U4 — p-<sup>11</sup>B plasma regime (physics, not a shelf material)

- **Headline channel:** <sup>1</sup>H + <sup>11</sup>B → 3 <sup>4</sup>He (+ energy)
- **Injectants:** H<sub>2</sub> + B<sub>2</sub>H<sub>6</sub> (dissociated in discharge) to tangential keV injectors
- **Power headline:** 3.5 MW gross thermal at full command (surrogate scale; model must derive conditions)
- **Beam:** > 1 mA class ion beam integration target
- **Density:** 10<sup>11</sup>–10<sup>12</sup> cm<sup>-3</sup> class goals

The simulator derives: required fusion power density, beam power, and confinement time — this is the **largest** unknown; U1–U3 must survive whatever power and wall load that solution demands.

### Validation levels and what each stage proves

- **0** — Pad interlock sequence — Correct startup order before fueling and reaction
- **1** — OD plant + U1–U4 gates — 3.5 MW headline, jet closure, materials limits
- **2** — Electron-ring simulation (stages 1–2) — Density and beam coupling at 600 kV — **not** fusion gain
- **3** — p-<sup>11</sup>B channel + burn models — ⟨σv⟩(T<sub>i</sub>) × fueling × volume; laminar / clump checks
- **4** — *Future* — Transport-integrated reactivity without analytical surrogate blend

**Critical honesty:** The electron-ring simulation integrates **electrons in prescribed E×B fields only**. It does not include fuel species, compressor airflow, or p-<sup>11</sup>B fusion yield. Fueling and the air-breathing Brayton path enter at later stages.

“first-principles proof of fusion”

All of the following in **one forward run** (steps 0→8, nominal reactivity assumption):

1.  $n_p$ ,  $n_B$ ,  $T_i$  from **PIC/transport**, not  $sccm \times \tau$  alone.
2. **No** surrogate blend or CSV calibration to hit MW.
3. Export documents **level 4** and external cross-check (literature or test-stand diagnostics).
4. (Hardware) measured power/products agree with steps 5–6 within uncertainty.

Until then, this chain is **progressive design refinement** toward that claim; step 9 documents unobtainium **margins** if the physics-only forward model misses 3.5 MW.

→ unobtainium components

- 0–1 — **U1** cathode gap / voltage / emission feasibility
- 2 — **U4** beam coupling (PIC deposit vs assumed mA)
- 3 — Laminar **design** (shear, pulse, injectant mix)
- 4–5 — **U4** fueling + reactivity path
- 6 — **U2** wall + CH<sub>4</sub>, **U3** HTS, gross power headline
- 7 — Plant  $\eta$ , thrust,  $\dot{m}$  consistency
- 8 — Design validation summary
- 9 — Minimum U1–U4 knobs if forward chain misses target

### Three-scenario benchmark

Target gross power **3.5 MW**.

- (a) Pretend — design target — design — 600 kV —  $1 \times$  — branch  $1e+03 \times$  @  $\eta=1$  — 3.228 — False
- (b) Today — literature  $\sigma_v$  + experimental anchors — literature — 300 kV —  $1 \times$  —  $1e+03 \times$  (branch $\times\eta$ ) — 1.852 — False
- (c) Minimum — stress inverse (literature  $\sigma_v$ ) — literature — 600 kV —  $751 \times$  — — (infeasible) — — — False
- Margin inverse @ design  $\sigma_v$  (back-solve check) — design — 600 kV —  $1.13 \times$  — back-solve vs (a) — 3.412 — True

*(b) today forward: 1.8518 MW (shortfall +1.65 MW vs target). At (a) voltage, design/literature  $\langle \sigma_v \rangle$  branch  $\approx 1000.0 \times$ .*

### Baseline at nominal Unobtainium

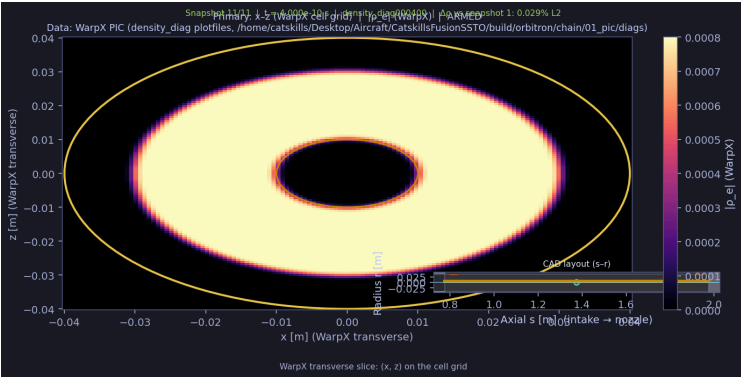
Nominal forward model at **design  $\sigma_v$**  and unity Unobtainium scales, targeting **3.5 MW** gross to the Brayton path.

- Layout — OK
- Electron ring — OK
- $p_e$  normalization —  $p_{e\_norm}=1.0000015855924544$
- Fusion channel — clump\_ON=1.05, OFF/ON=2.50 $\times$ , armed=True
- Fueling — OK
- p-<sup>11</sup>B burn —  $P_{fusion}=3.58$  MW
- 0D plant + U1–U4 —  $P_{gross}=3.58$  MW, Feasible=True
- Jet closure — closure=0.00%

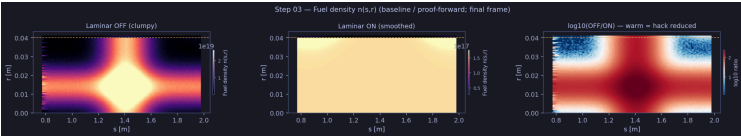
- Validation — design closure=yes

Plasma channel and fusion fields

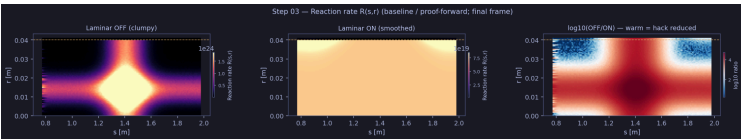
The electron ring sets the density scale that feeds the laminar fusion-channel model. With fueling armed, reaction-rate structure and clump metrics show how much power couples into the volume before the 0D burn step.



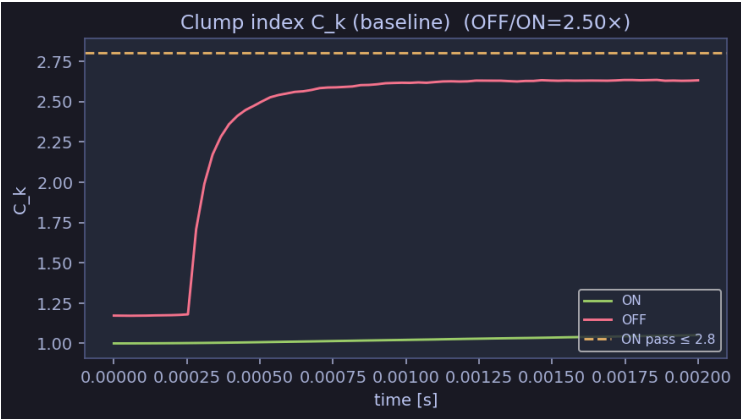
Step 1. WarpX. rho\_e (last)



Step 3. Final density



Step 3. Final reaction rate

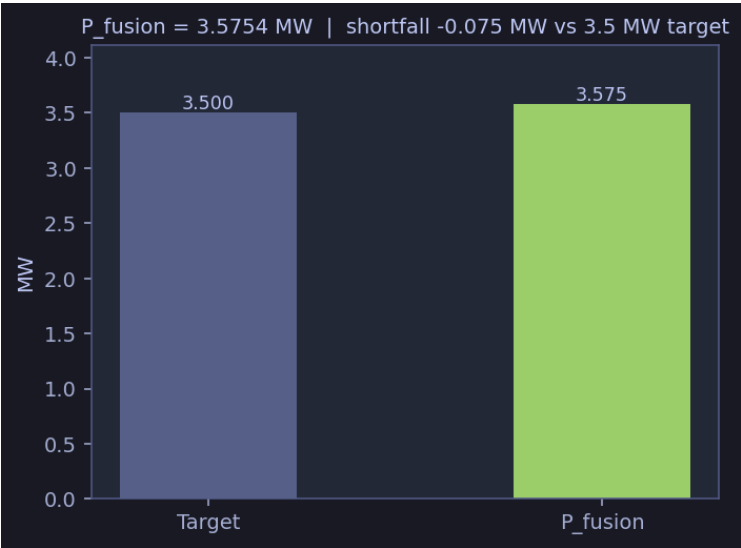


Step 3. Clump index

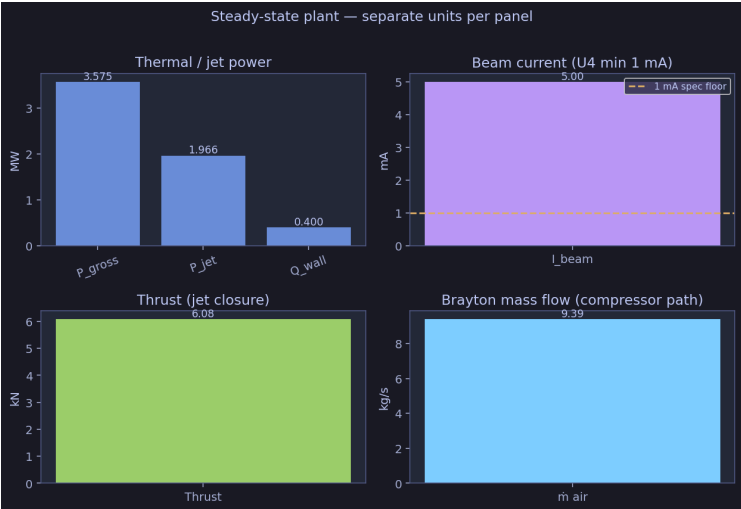
Integrated channel power **3.575 MW**; clump OFF/ON **2.50×**.

Burn power and plant closure

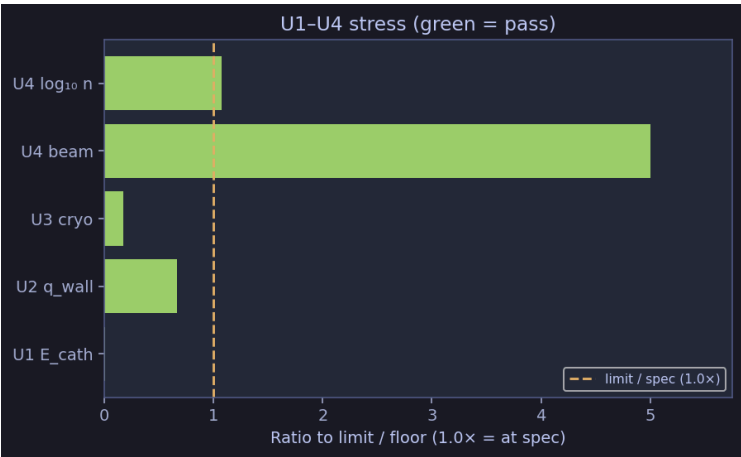
p-11B burn power **3.575 MW** against the gross target. 0D plant gross **3.575 MW**, Feasible **True**.



Step 5. Burn power



Step 6. Plant outputs



Step 6. Unobtainium stress

Thrust path

Jet closure relative error **0.00%**. Booked thrust **1366.1 lbf** on the thrust sled.



Step 7. Jet closure

### Honest gap to 3.5 MW (constrained stress inverse)

Under **literature-class**  $\langle\sigma v\rangle$  we minimize  $\eta_{\text{react}}$  (and other knobs) with a **trust-region constrained** solve: power  $\geq \{\text{target:g}\}$  MW and **U1–U4** are hard inequalities. If no feasible point exists, scenario **(c)** is **infeasible** — not a violation-bearing “solution.”

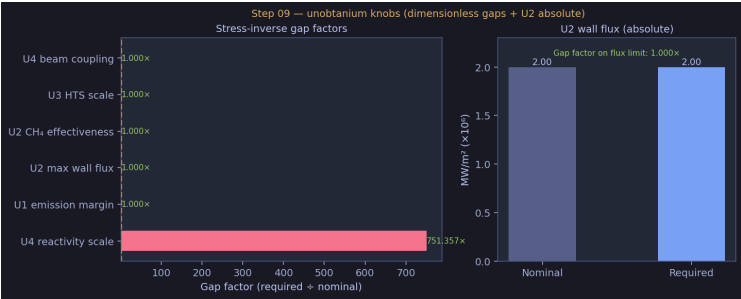
Primary reactivity gap at solve:  $\langle\sigma v\rangle$  design/literature  $\approx 1000.0\times$ ;  
 fusion\_reactivity\_scale required  $\approx 751.4\times$ ; combined  $\approx 751356.1\times$  vs nominal scale.

**Constrained stress inverse: INFEASIBLE** — no literature- $\sigma v$  point meets **3.5 MW** and **U1–U4** together. The  $\langle\sigma v\rangle$  branch gap ( $\sim 10^3\times$ ) cannot be closed with the allowed knob bounds while satisfying cathode, wall, and magnet gates. Scenario **(c)** in the table above is marked infeasible; do not treat optimizer knobs as a operating solution.

*Optimizer best-effort knobs (gates not satisfied — illustrative only):*

- beam coupling scale — **1.000×**
- ch4 cooling effectiveness — **1.000×**
- field emission margin — **1.000×**
- fusion reactivity scale — **751.356×**
- hts capability scale — **1.000×**
- max wall heat flux W m<sup>2</sup> — **1.000×**

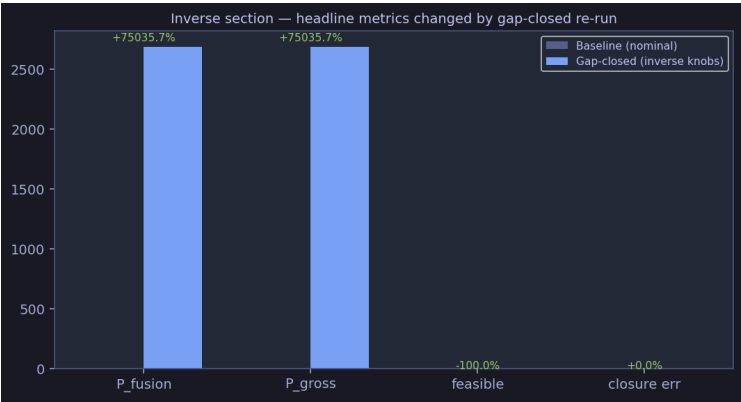
Forward check at **design**  $\sigma v$  with solved knobs: **PASS (3.344 MW)**.  
 Uses **margin-inverse** knobs on design  $\sigma v$  (back-solve  $\approx$  pretend), not stress-inverse  $\eta_{\text{react}} \sim 10^3\times$  on literature  $\sigma v$ .



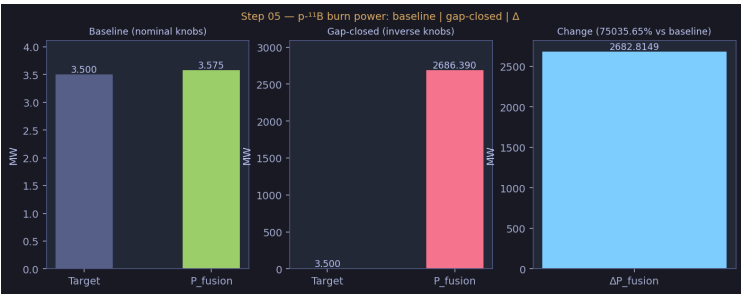
Step 9. Unobtainium comparison

Performance at gap-closed Unobtainium

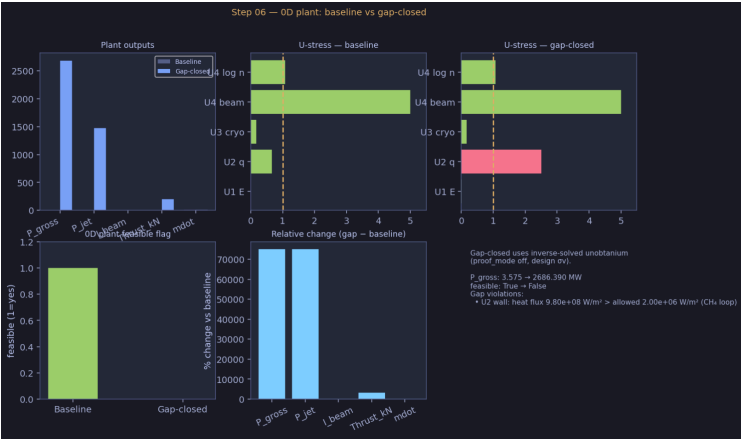
Re-running the fusion channel and plant with **inverse-solved** Unobtainium knobs (design σv, literature stress basis). Compare to baseline panels below.



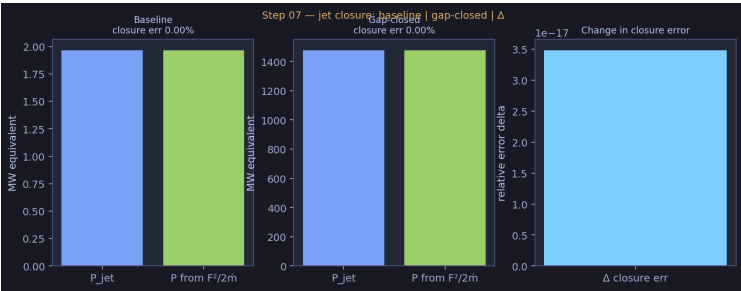
Invert model, summary comparison



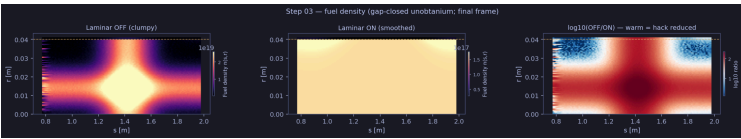
Step 5. Burn comparison



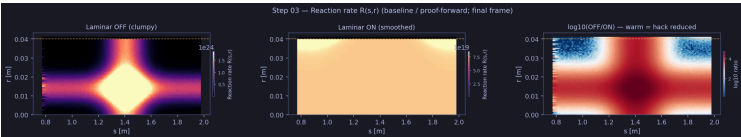
Step 6. Plant comparison



Step 7. Closure comparison



Step 3. Gap final density



Step 3. Gap final reaction rate

Gap-closed channel power **2686.386 MW**. Gap-closed gross **2686.386 MW**. First-tier gates at gap-closed knobs: **no**.

### Conclusion — technology gaps and R&D program

**Experiment:** Orbitron Direct Cycle p-11B 3.5 MW Turbojet In Silico Benchmark **Geometry (concept):** coaxial bore — cathode radius ~10 mm, anode/plasma radius ~40 mm, active length 1.2 m; design cathode potential ~600 kV, axial field 2.0 T in bore. **Fueling (held across scenarios):** H<sub>2</sub> ~135 sccm equivalent + laser ablation ~16.9 Hz on <sup>11</sup>B target index 0 (B<sub>2</sub>H<sub>6</sub> dissociation path in discharge).

**Main result:** On literature (σv), there is **no feasible constrained minimum** for 3.5 MW gross thermal while satisfying U1–U4 and power closure. Constrained stress-inverse success is **False**. The plant model therefore does not identify a literature-grounded operating point; it reports an effective (σv) design/literature branch ≈ 10<sup>3</sup> (reported

$\sim 1000\times$ ) and a best-effort  $\eta_{\text{react}} \approx 7.5 \times 10^2$  — i.e. fusion reactivity would need to be enhanced by **roughly three orders of magnitude** relative to tabulated thermonuclear reactivity at the assumed conditions, not tuned by a few percent in wall alloy or cryostat margin.

(a) Pretend closes at the **design-calibrated  $\langle\sigma v\rangle$**  and  $\sim 600$  kV with level-1 validation and margin confirmation **PASS** at  **$\sim 3.34$  MW** — this is **internal design-basis consistency**, not a claim that the reactor has demonstrated fusion power.

(b) Today at **Avalanche-class  $\sim 300$  kV**, literature  $\langle\sigma v\rangle$ , and experimentally bounded wall/HTS limits with the **same fueling as (a)**, does not close 3.5 MW without the design  $\sigma v$  branch.

(c) Minimum is **infeasible** on the stated constraints. **Overall R&D likelihood of reaching 3.5 MW on literature physics with this geometry and fueling: low.**

Electron-loading physics has been checked at **validation level 2** (particle-in-cell class studies); that tier does **not** validate fusion gain, Q, or net thermal power.

## Scenario interpretation

### (a) Pretend — design-calibrated $\langle\sigma v\rangle$ , $\sim 600$ kV (level-1 plant closure)

Scenario (a) assumes a **design-point reactivity  $\langle\sigma v\rangle$**  calibrated to close the 0D energy and materials budget at  **$\sim 600$  kV, 2 T**, and the stated injectants. Level-1 design validation is **True**: U1–U4, wall intercept, cryo leak, and Brayton-side bookkeeping are mutually consistent at the  **$\sim 400$  kW** first-wall anchor and **3.5 MW** command scale.

**Margin confirmation at design  $\sigma v$**  passes at  **$\sim 3.34$  MW** with residual power margin  **$\sim +0.61$  MW** relative to the closure target (negative residual in the stress run denotes shortfall on the literature branch). Interpreting (a) correctly: it demonstrates that **the spreadsheet/plant model is self-consistent given its input  $\sigma v$** , analogous to a design-basis heat balance that assumes a catalog pump curve — **not** that measured p- $^{11}\text{B}$  yield at 600 kV has been demonstrated at MW scale.

### (b) Today — literature $\langle\sigma v\rangle$ , $\sim 300$ kV “Avalanche” class, experimental limits

Scenario (b) swaps design  $\sigma v$  for **published thermonuclear  $\langle\sigma v\rangle$**  (and related cross-section compilations), reduces the electrostatic scale to  **$\sim 300$  kV** consistent with an Avalanche-class operating picture, and applies **experimental** cathode-field, first-wall, and HTS/cryostat limits. Fueling is unchanged from (a).

At 300 kV the beam–plasma interaction moves away from the design reactivity peak in energy space; even **recent compilations** that raise  $\langle\sigma v\rangle$  by  **$\sim 20$ – $50\%$**  near **200–300 keV** ion temperatures (Sikora–Weller / Tentori–Belloni line; Mazzucconi et al. 2025 resonance structure) are **orders of magnitude smaller** than the  **$\sim 10^3$**  branch required here. (b) should be read as: **with today’s tabulated nuclear data and today’s voltage/material caps, the declared mission power is not carried.**

### (c) Minimum — constrained stress inverse on literature $\langle\sigma v\rangle$

The constrained optimizer seeks the **smallest  $\eta_{\text{react}}$**  (effective reactivity multiplier vs literature) subject to U1–U4, **3.5 MW**, and power balance. **Success: False.** No point satisfies literature  $\langle\sigma v\rangle$ , full power, and unit constraints simultaneously.

Best-effort  $\eta_{\text{react}} \approx 751$  is a **mathematical lower bound on the multiplier still insufficient for feasibility** — not a validated operating mode. **Feasible minimum: none.**

### Dominant gap — $\langle\sigma v\rangle$ branch vs materials

- **$\langle\sigma v\rangle$  design / literature —  $\sim 10^3$  — Dominant.** Closes (a); breaks (b) and (c).
- Updated  $\sigma(E)$  literature (2023–2026) —  **$\sim 1.2\text{--}1.5\times$**  — Does not explain the branch.
- $\eta_{\text{react}}$  (optimizer best effort) —  **$\sim 7.5\times 10^2$**  — Same physics gap expressed as required enhancement factor.
- Material / thermal knobs (U2–U3) —  **$\sim \text{few \% to tens \%}$**  — Cannot absorb a  $10^3$  reactivity deficit; stress inverse never finds a feasible material-only minimum.

**Interpretation for specialists:** The benchmark's binding constraint is **not** whether C-103 can hold **800–1000 °C** or whether YBCO at  **$\sim 113\text{ K}$**  survives parasitic leak — those matter for **survivability** once fusion power is specified — but **whether the specified fueling and geometry produce enough fusion power density at literature  $\langle\sigma v\rangle$** . The model answers **no** without a reactivity input  **$\sim 1000\times$**  above literature at the relevant effective temperature/energy.

Recent work improves **absolute  $\langle\sigma v\rangle$**  modestly and revisits **bremsstrahlung vs fusion** balance with updated  $\sigma$  and **Ti/Te** (e.g. Frontiers 2024 avalanche-style scenarios; 2025–2026 energy-balance preprints). None of that collapses a  $10^3$  gap to a credible near-term experiment.

### Knob-by-knob (materials)

**No feasible (c)** — a knob table for wall alloy,  $\text{CH}_4$  intercept fraction, MLI thickness, or HTS current margin would **misleadingly imply** a tunable minimum exists. Infeasibility is driven by  **$\eta_{\text{react}} \sim 10^3$** , not by U2 niobium grade or U3 cryostat parasitics at the  **$\sim \text{few-percent}$**  level.

Qualitatively: if literature  $\langle\sigma v\rangle$  were magically satisfied, **U2** ( $\alpha/\text{X-ray}/\text{CX}$  on first wall,  **$\sim 55\%$**  of wall load to liquid  **$\text{CH}_4$**  channels) and **U3** (HTS **outside** vacuum cryostat, radiative leak only) set **survivability** bands; they do **not** substitute for fusion power. **U1** ( $\sim 600\text{ kV}$ , field emission vs gap) and **U4** (density  **$10^{11}\text{--}10^{12}\text{ cm}^{-3}$** ,  **$>1\text{ mA}$**  class beam integration) remain coupled unknowns in any forward path.

### Recommended R&D program (ordered)

Measurements that would **collapse uncertainty** on the actual bottleneck (reactivity and power density), then thermal integration:

1. **Absolute p- $^{11}\text{B}$  reactivity at relevant mean energies** — time-resolved  $\sigma(E)$  or yield vs  **$E_p$ ,  $E_B$** , and composition; compare to design-branch implied  $\langle\sigma v\rangle$  at same  **$n_p$ ,  $n_B$** , and effective **T** (not Maxwellian peak alone).
2. **Beam-target / laser-ablation campaign at fixed geometry** (40 mm / 10 mm / 1.2 m scale model) —  $\alpha$  yield, neutron-free power

accounting, and **>1 mA** equivalent ion current with  $H_2 + B$  injection as in benchmark fueling.

3. **Voltage sweep 300–600 kV with arc-free U1 metrics** (emission current density, breakdown statistics) — separates electrostatic feasibility from reactivity shortfall.
4. **In-situ density ( $10^{11}$ – $10^{12} \text{ cm}^{-3}$ ) and Ti/Te** — Thomson, spectroscopy, or passive constraints; test whether non-Maxwellian or avalanche tails can exceed literature  $\langle\sigma v\rangle$  by **orders** (not  $\sim 30\%$ ).
5. **WarpX / PIC level 2 to level 3** — electron loading and field structure at **2 T**; extend only after (1)–(4) bound **fusion** relevant quantities (PIC alone does not prove gain).
6. **First-wall mockup** — **C-103-class** surface at **800–1000 °C** with **liquid  $CH_4$**  channel intercept of  $\sim 55\%$  of  $\sim 400 \text{ kW}$  class deposition; measure  $CH_4$  flux closure **independent** of Brayton air.
7. **Air annulus + first-wall zoning** — hot air flow **inside** the solenoid bore, over the first wall; measure  **$\alpha$ /X-ray** heated air outlet state vs turbine inlet requirement; identify need for **liquid-to-air HX** if annulus alone is insufficient.
8. **Cryostat + MLI integration test** — HTS **outside** vacuum cryostat; hot annulus separated by vacuum gap + MLI; quantify radiative leak to  $\sim 113 \text{ K}$   $CH_4$  loop and **refrigeration power** (OD does not size cryoplant mass).
9. **HTS solenoid 2 T** in bore with combined **thermal + EM** cycle — confirm **no** direct charged-particle flux on tape; parasitic only.
10. **Closed-loop Brayton segment** — air heated by annulus only; pad/thrust bookkeeping without Category-E multi-MV DEC assumptions; measure net shaft power vs **any** measured fusion thermal (if ever available).

## Risks & unknowns (OD plant fidelity)

### Likely overstated

- **Fusion power** at design  $\sigma v$  — closure is **by construction** in (a).
- **Bremsstrahlung and Q** — updated  $\sigma$  slightly **favors** p- $^{11}\text{B}$  balance in some models, but not  $10^3$ .
- **Simplicity of OD beam-target** — real keV injectors, dissociation, and recycling may **lower** effective  $\langle\sigma v\rangle$  vs ideal beam-target.

### Likely understated

- **Cryoplant mass and refrigeration power** —  $CH_4$  at  $\sim 113 \text{ K}$  for HTS plus intercept loop; SSTO mass not closed in OD.
- **Liquid-to-air heat exchange** — if annulus air cannot reach Brayton inlet temperature, extra HX surface and pressure drop appear.
- **$CH_4$  chemistry and carbon transport** on hot wall — not fully coupled to plasma impurity limits.
- **Field emission and micro-arcing** at  $\sim 600 \text{ kV}$  — U1 may force lower voltage before U4 fails.
- **PIC tier** — level 2 validates **electrons**, not **ion fusion** or  **$\alpha$  heating** feedback.
- **Removed Category-E paths** (+4 MV DEC, diamond MV inverters) — if needed electrically, **air Brayton-only** bookkeeping is optimistic for pad energy.

**Thermal layout (correct zoning):** The HTS solenoid sits **outside** the vacuum cryostat. **Hot air** flows in an **annulus inside the magnet**, over

the **first wall** carrying  $\alpha$  and X-ray load. **Liquid CH<sub>4</sub>** in wall channels intercepts a large fraction of wall heat; **heated air** feeds the Brayton cycle. Air does **not** cool the HTS; the magnet is **not** the Brayton jacket.

## Conclusions

For a skeptical nuclear/plasma reader, the benchmark's scientific content on the **literature path** is blunt: **3.5 MW thermal closure with this geometry, fueling, and U1–U4 is infeasible without reactivity  $\sim 10^3$  above tabulated p-<sup>11</sup>B ( $\sigma_v$ )**. Recent cross-section work improves reactivity by **tens of percent**, not **three orders of magnitude**. Scenario (a) is a **design-basis consistency check** at calibrated  $\sigma_v$ ; margin confirmation  **$\sim 3.34$  MW** does **not** constitute a physical fusion-power demonstration. Scenario (b) represents a more credible near-term voltage and materials posture and still fails power closure. Scenario (c) does not exist as a feasible constrained minimum.

**Validation status:** Electron-loading studies (e.g. WarpX-class) support **electrostatic structure and loading** at level 2; they **do not** prove fusion gain or net reactor power. **Overall likelihood** of the stated 3.5 MW mission on literature physics without fundamental changes to confinement, effective energy, or reactivity assumptions: **low**. The prioritized program should measure ( $\sigma_v$ ) and **power density** first; thermal and cryostat integration experiments remain necessary but **secondary** until the reactivity gap is addressed with facility-grade data.

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